

Master Thesis Supervisor Meeting – Briefing Agenda

Mode-I Phase-Field Adaptive Remeshing Benchmark Reproduction

Date: Thursday, 17 September 2026 · **Duration:** 45 Minutes

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Governing Supervisor Directive

“We need to have understood everything related to the first model before we increase complexity.”

Meeting Agenda & Discussion Flow (45 Minutes)

1. Mode-I Benchmark Definition & Fixed-Mesh Reference Anchor (00–08 min)

- Single-edge-notched square plate (1.0×1.0 mm) with zero-gap sharp seam ($a_0 = 0.5$ mm).
- Fixed-mesh reference anchor (Job 1398090, 15,192 elements): $F_{\max} = 0.757778$ kN at $u = 0.005857$ mm; authoritative unconstrained OLS structural stiffness $K_0 = 137.945520$ kN/mm ($R^2 = 0.99999960$).
- Multi-quantity convergence criteria and UEL internal energy balance limitation note.

2. Pandey–Kumar Native Adaptive Workflow & Error Indicator (08–15 min)

- Automated two-job procedure: coarse linear elastic pre-analysis → MISESeri extraction → native `adaptiveRemesh` → dual UEL/UMAT mesh generation → nonlinear fracture solve.
- Epistemic clarity on MISESeri: Abaqus stress discretization error indicator associated with recovered von Mises stresses in linear elasticity; not a phase-field or damage indicator.
- Canonical Mode-I pre-analysis provenance: 2,906 finite elements, 2,906 `WHOLE_ELEMENT` scalar records (formal retirement of obsolete Mode-II 3,930-row dataset).

3. Resolution of the 71,320-Element Stiffness Defect (Priority Question A) (15–25 min)

- Root-cause isolation: Abaqus `pre` 16-node card limit on free-format `*NSET` records without `GENERATE`. Abaqus issued warnings and retained only 16 of the 150 `N_BOTTOM` nodes; 134 entries were deleted.
- Kinematic mechanism: unconstrained nodes permitted bottom-edge vertical lift up to $\approx 48.34\%$ of applied stroke, artificially softening structural stiffness down to $K_0 \approx 122.38$ kN/mm.
- Verified correction: wrapping records to ≤ 16 entries/line constrained all 150 bottom nodes to $u_2 = 0$, recovering exact stiffness: $K_0 = 138.021013$ kN/mm (frozen, $+0.05\%$) and $K_0 \approx 137.821$ kN/mm (137.820804 kN/mm, full fracture Job 1404933, -0.09%).

4. Discrepancy Audit: 71,320 vs $\approx 13,941$ Finite Elements (Priority Question B) (25–35 min)

- Multi-release invariance: Abaqus 2019 GA through 2023.HF4 generate 100.000% bitwise identical 71,320-element meshes with identical SHA-256 hashes.
- Exhaustive 15-factor OFAT audit: ruled out release shifts, load scale, output frequency, mesher controls, and multi-pass remeshing; tested boundary conditions and element formulations.
- Regional distribution: crack process corridor ($|y| \leq 0.05$ mm) contains 9,061 elements (12.7%), while far-field and transition account for 62,259 elements (87.3%) of our reconstruction.
- Literature gap: accessible evidence does not identify which unpublished implementation detail accounts for the reported $\approx 13,941$ mesh; author input is required for definitive closure.

5. Supervisor Decisions Required & Next Steps (35–45 min)

- **Option A (Recommended):** Accept Gate 5 as externally under-specified, retain 71,320 as the publication-literal reconstruction, document $\approx 13,941$ as an unresolvable literature gap, and proceed with Mode-I thesis synthesis/documentation.
- **Option B:** Authorize transmission of the prepared 6-question author inquiry and keep Gate 5 open.
- **Explicit Hold:** Under both options, Gate 7 (`ABAQUSER`), Mode-II, and state transfer remain on hold.